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(Pages : 3) Name.....

Reg. No.....

FIRST SEMESTER B.Sc. DEGREE EXAMINATION, JANUARY 2012

(CCSS)

Physics—Complementary

PH 1C 01—PROPERTIES OF MATTER AND THERMODYNAMICS

ime : Three Hours Maximum Weightage : 30

Answer **all** *twelve questions.*

I. Choose the correct alternative :

1 Which of the following is a correct relationship between various elastic constants (Young's modulus (Y), Rigidity modulus (n) Bulk modulus (K) and Poisson's ratio (σ) ?

(a) $Y = \frac{K}{3(1-2\sigma)}$. (b) $\frac{1}{Y} = \frac{1}{3n} + \frac{1}{9K}$.

(c) $\sigma = \frac{(3K+2n)}{(3K-2n)}$. (d) $n = \frac{2(1+\sigma)}{Y}$.

2 The excess pressure in side an air bubble of radius r well within a liquid of surface tension T is :

(a) $4T/r$. (b) T/r.

(c) $2T/r$. (d) $r/4T$.

3 The slope of the P–V diagram (V along X–axis and P along Y–axis) of an isochoric process is :

(a) zero. (b) infinity.

(c) P/V. (d) P.

4 Enthalpy H =

(a) U–PV. (b) A + TS.

(c) $U^2 + PV$. (d) None of the above.

Fill in the Blanks :

5 When thickness of a rectangular beam is doubled, Bending Moment becomes ———— times.

6 When sodium chloride is dissolved in water surface tension ————.

7 When the volume of an ideal gas is doubled at constant temperature, work done = ————.

8 Change in Entropy in a ———— cycle is zero.

Turn over

Give *one* word answers :

9 What is the equation of motion of a torsion pendulum ?

10 What is the unit of coefficient of viscosity ?

11 What is the value of universal gas constant ?

12 Which law of thermodynamics can be used to explain the working of a refrigerator ?

(12 × ¼ = 3 weight

Answer **all** *nine questions.*

II. 13 What is the reason for I–cross section for girders ?

14 Distinguish between Uniform and Non–Uniform bending.

15 A torsion pendulum of length 25 cm has a period 10 seconds. Find its period when len is increased to 50 cm.

16 How does Surface Tension vary with temperature ?

17 What is Brownian motion ?

18 From the T–S diagram of Carnot cycle derive an equation for its efficiency.

19 Derive an equation for work done in Isobaric Process.

20 What is the Principle of increase of entropy ?

21 Write down Clausius–Clapeyron equation.

(9 × 1 = 9 weighta

Answer any **five** *questions.*

III. 22 Derive an equation for bending moment for a uniformly bent beam.

23 Define Poisson's ratio. Derive the theoretical limiting values of Poisson's ratio.

24 A number of little droplets of water of radius *r* each coalesce to form a single drop of rad R. Show that rise in temperature of water is $\frac{3S}{c\rho}\left(\frac{1}{r}-\frac{1}{R}\right)$ where S is the surface tension the specific heat capacity and ρ is the density.

25 Calculate the work done in blowing a soap bubble of radius 7 cm and surface tension ($N\,m^{-1}$.

26 Two equal drops of water are falling through air with a steady velocity of 2 cm/second the drops coalesce find the new terminal velocity.

27 An ideal gas at S.T.P. is first compressed to ¼ th of its volume adiabatically and th expanded to its initial volume isothermally. Calculate the net amount of work done (γ = 1.

28 State and prove Carnot's Theorem.

(5 × 2 = 10 weighta

Answer any **two** *questions.*

29 Define a Cantilever. With necessary theory explain how a cantilever can be used to find the Young's Modulus.

30 Derive Poiseuille's formula.

31 Explain thermodynamic potentials. Derive Maxwell's thermodynamic relations.

(2 × 4 = 8 weightage)